

← Match found      + Added <instrument, verb, target>  
? Unable to match automatically      - Removed <instrument, verb, target>  
 + Instrument-ID      Added Instrument mask with Instance ID

(1) CholecT50 CholecInstanceSeg

**Step 1**  
Automatic Matching fails  
(Missing Triplet)

Grasper - 1      Hook-1

<Grasper, Grasp, Gallbladder>

?

**Step 2**  
Add Missing Triplet Label

Grasper, Grasp, Gallbladder      Hook-1

<Grasper, Grasp, Gallbladder>

+ <Hook, Dissect, Gallbladder>

**Final Result**

Grasper, Grasp, Gallbladder      Hook, Dissect, Gallbladder

(2) CholecT50 CholecInstanceSeg

**Step 1**  
Automatic Matching fails (More  
Triplets than Instances)

Grasper - 1      Hook-1

<Grasper, Null\_Verb, Null\_Target>

? <Hook, Null\_Verb, Null\_Target>

**Step 2**  
Delete Extra triplet

Grasper, Null\_Verb, Null\_Target      Hook-1

<Grasper, Null\_Verb, Null\_Target>

- <Hook, Null\_Verb, Null\_Target>

**Final Result**

Grasper, Null\_Verb, Null\_Target      Hook, Null\_Verb, Null\_Target

(3) CholecT50 CholecInstanceSeg

**Step 1**  
Automatic Matching fails  
(Missing instrument instance)

Grasper - 1      Hook-1

<Grasper, Retract, Gallbladder>

<Hook, Retract, Gallbladder>

**Step 2**  
Add missing instrument Instance

Grasper - 1      Hook, Retract, Gallbladder

<Grasper, Retract, Gallbladder>

+ <Grasper, Retract, Gallbladder>

**Final Result**

Grasper, Retract, Gallbladder      Hook, Retract, Gallbladder